

Real-time Vision-Based Hand Gesture Recognition System for Emergency Sign Language Interpretation

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Abstract—Deaf and hard-of-hearing (DHH) individuals frequently encounter critical communication barriers in emergency situations where sign language interpreters are unavailable. This paper presents a real-time vision-based hand gesture recognition system that translates emergency sign language gestures into phrases for hearing individuals without requiring specialist hardware. The system employs Google MediaPipe for hand landmark extraction, feeding an 84-dimensional feature vector into a lightweight Feedforward Neural Network (FNN) classifier. A custom dataset, EmergencySL, comprising 60,000 samples across 15 emergency gesture classes, was collected under varied real-world conditions. Through iterative architectural experimentation, the final model achieved a test accuracy of 95.46% and a macro F1-score of 0.96 across all 15 classes. The system was deployed as a Streamlit web application with real-time WebRTC webcam streaming and text-to-speech output, running entirely on standard consumer hardware without GPU acceleration. Results demonstrate that lightweight landmark-based classification is a viable and accessible approach to emergency sign language interpretation.

Index Terms—sign language recognition, hand gesture recognition, MediaPipe, feedforward neural network, emergency communication, accessibility, real-time inference, EmergencySL

I. INTRODUCTION

Deaf and hard-of-hearing (DHH) individuals rely primarily on sign language as their main form of communication. However, most hearing individuals, including emergency responders, lack proficiency in sign language, creating significant communication barriers in time-critical emergency situations. McKee et al. [1] demonstrated that DHH individuals frequently experience delays in receiving emergency aid due to these barriers. While communication services such as British Sign Language (BSL) interpreters, Video Relay Services, and the 999 BSL emergency video service exist, Shank and Foltz [2] found that emergency BSL interpreters in the UK are rarely available outside regular hours, with interpreters typically fully booked weeks or months in advance.

Advancements in computer vision and machine learning present an opportunity to address this gap. Hand landmark detection tools such as Google MediaPipe [3] can extract precise keypoint coordinates from live video streams using standard cameras, while lightweight neural network classifiers can learn to recognise gesture patterns from these coordinates in real time.

This paper makes the following contributions:

- Introduction of **EmergencySL**, a custom dataset of 60,000 samples across 15 emergency sign language gesture classes collected under varied real-world conditions.
- A lightweight FNN-based gesture classification system achieving 95.46% test accuracy, deployable on commodity hardware without a GPU.
- A complete end-to-end real-time web application integrating gesture recognition with emergency phrase mapping and text-to-speech output.

The project repository is available at: <https://github.com/Samm-006/Hand-Gesture-Recognition.git>

II. RELATED WORK

A. Sign Language Recognition

Sign language recognition (SLR) has evolved from early sensor-based approaches requiring specialised hardware such as data gloves to vision-based methods using standard cameras. Cooper et al. [4] provide a comprehensive survey of SLR techniques, noting that publicly available datasets remain limited in quality and quantity. Cheok et al. [5] identify vision-based approaches using single cameras as the dominant research direction, with deep learning models such as CNNs and RNNs achieving state-of-the-art accuracy. However, these models are computationally demanding and require large datasets, limiting their practicality for real-time deployment on non-specialised hardware.

B. MediaPipe-Based Gesture Recognition

Google MediaPipe [3] provides a hand landmark detection pipeline trained on approximately 30,000 real-world images and synthetic hand models. Bora et al. [6] demonstrated that combining MediaPipe hand keypoints with a deep learning classifier enables fast and accurate sign language recognition. Zhang et al. [3] showed that using landmark coordinates as input features significantly reduces computational cost while maintaining strong classification performance compared to raw image-based approaches.

C. Emergency Sign Language Systems

Despite advances in SLR, systems specifically designed for emergency sign language communication remain scarce. Existing systems prioritise large general vocabularies over the reliability and speed required in emergency scenarios. A model

trained on general sign language vocabulary cannot reliably classify emergency-specific gestures absent from its training data, a critical limitation when every gesture must be correctly identified.

III. SYSTEM DESIGN

A. Pipeline Overview

The system operates as a four-stage pipeline as illustrated in Fig. 1: (1) hand detection and landmark extraction using MediaPipe, (2) preprocessing and feature extraction, (3) FNN gesture classification, and (4) real-time inference and web application deployment. The modular design allows each stage to be developed and tested independently.

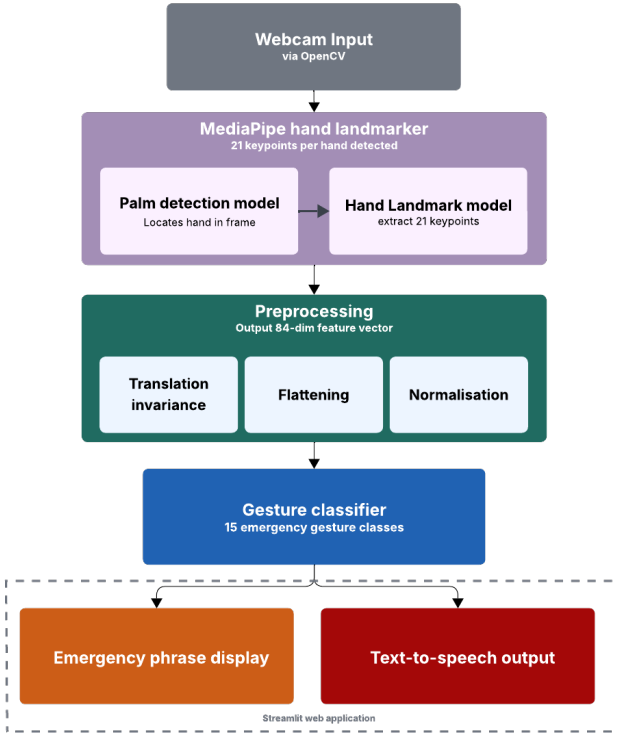


Fig. 1. End-to-end system architecture pipeline.

B. Hand Detection and Landmark Extraction

MediaPipe Hand Landmarker is configured in LIVE_STREAM mode, processing frames asynchronously via a callback function to maintain low-latency performance. The landmarker detects up to two hands simultaneously with detection and tracking confidence thresholds set to 0.5, the default recommended value [7]. For each detected hand, 21 keypoint (x, y) coordinates are extracted as illustrated in Fig. 2, yielding a 42-dimensional vector per hand and 84 dimensions for two hands.

Webcam frames are flipped horizontally before processing, as MediaPipe assumes horizontally flipped input when determining handedness [7]. Frames are converted from BGR to RGB format as required by MediaPipe.

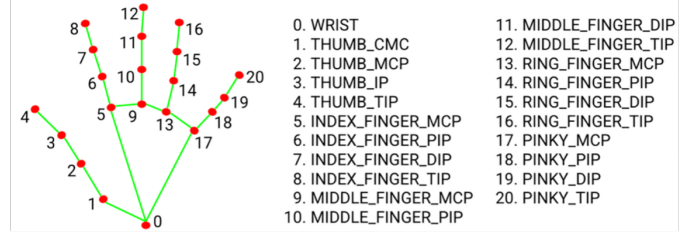


Fig. 2. MediaPipe hand landmark keypoints (0–20).

C. Preprocessing

Three preprocessing steps are applied before classification:

- 1) **Translation invariance:** Wrist coordinates (keypoint 0) are subtracted from all other landmark coordinates, ensuring the gesture representation is independent of hand position in the frame.
- 2) **Flattening:** The 2D landmark array is converted into a flat 1D vector of 42 values per hand.
- 3) **Normalisation:** Each value is divided by the maximum absolute value of the vector, providing scale invariance across different hand sizes and camera distances.

The same preprocessing pipeline is applied identically during both data collection and inference, ensuring consistency between training and deployment.

IV. EMERGENCYSL DATASET

No existing public sign language dataset covers the specific emergency gesture vocabulary required for this system. WLASL [13], the largest video-based ASL dataset, stores raw video footage rather than landmark coordinates, requiring full reprocessing through MediaPipe and introducing inconsistencies in signing style and recording conditions. Image-based datasets such as the ASL Alphabet Dataset cover only finger-spelling, which is impractical for emergency communication where spelling terms letter by letter is too slow.

To address this gap, **EmergencySL** was introduced — a custom dataset of 60,000 samples across 15 emergency gesture classes using a standard RGB webcam and the same MediaPipe pipeline used for inference. The 15 gesture classes were selected based on Engel [11] and Shank and Foltz [2], and are listed in Table I.

TABLE I
EMERGENCYSL GESTURE VOCABULARY (15 CLASSES, LABELS 0–14)

Label	Gesture	Label	Gesture
0	I	8	Police
1	Need	9	Fireman
2	Accident	10	Hurt
3	Fire	11	Emergency
4	Yes	12	Short of breath
5	No	13	Help
6	Ambulance	14	Stop
7	Doctor		

Data was collected under varied real-world conditions to improve generalisation, as summarised in Table II. Approximately 4,000 samples were collected per gesture class across multiple sessions of 250 frames each, captured at 25–30 frames per second. EmergencySL has the potential to support future research in emergency sign language recognition beyond the scope of this project.

TABLE II
EMERGENCYSL DATASET COLLECTION CONDITIONS

Condition	Variations
Lighting	Good lighting, Poor lighting
Hand orientation	Straight-facing, Side-oriented
Camera distance	~50cm (close), ~120cm (far)
Hand dominance	Right hand, Left hand

V. MODEL

A. Architecture Selection

A Feedforward Neural Network was selected as the most appropriate classifier for this task. CNNs offer no architectural advantage over a flat landmark coordinate vector as input [8]. RNNs and LSTMs require sequentially structured data, whereas EmergencySL stores independent frames without temporal ordering [12]. FNNs process flat input vectors directly, require no special data structure, and are simpler to implement and train.

B. Network Architecture

The final architecture was determined through iterative experimentation and is summarised in Table III.

TABLE III
FINAL FNN ARCHITECTURE

Layer	Configuration
Input	84 features
Dropout	rate 0.1
Dense	256 units, ReLU
Dropout	rate 0.2
Dense	128 units, ReLU
Dropout	rate 0.1
Dense	64 units, ReLU
Output	15 units, Softmax
Total parameters: 63,887	

The pyramidal layer structure (256→128→64) encourages increasingly abstract feature representations at each layer [8]. ReLU activation was selected for all hidden layers due to computational efficiency and avoidance of the vanishing gradient problem [8].

C. Training Configuration

The Adam optimiser [9] was used with sparse categorical cross-entropy loss, as class labels are stored as integers rather than one-hot encoded vectors. The dataset was split 80/20 using stratified sampling, yielding 47,999 training samples

and 12,000 test samples with exactly 800 samples per class. Training used a maximum of 300 epochs with a batch size of 128. Early stopping monitored validation loss with a patience of 20 epochs, triggering at epoch 130 with a best validation loss of 0.1416.

VI. SYSTEM IMPLEMENTATION

A. Framework and Deployment

The trained model was deployed as a Streamlit web application [10], selected for its rapid development capability and compatibility with Python-based machine learning libraries. Real-time webcam streaming is enabled via the streamlit-webrtc library using a STUN server configuration (stun.l.google.com:19302). The TensorFlow model and gesture labels are loaded once at startup using Streamlit's `@st.cache_resource` decorator to ensure smooth real-time performance.

The deployed system maintained stable, low-latency inference on standard consumer hardware without requiring GPU acceleration, confirming the system meets its real-time deployment requirement on CPU-based systems.

B. Video Processing and Inference Pipeline

Each webcam frame is handled by a `VideoProcessor` class implementing a `recv()` method. For each frame, the system flips horizontally for correct handedness detection [7], converts BGR to RGB, extracts hand landmarks via `MediaPipe`, preprocesses into an 84-dimensional feature vector, and classifies with the trained FNN.

The `VideoProcessor` runs in a separate thread from the Streamlit interface. Two thread locks prevent race conditions: `_result_lock` protects the latest `MediaPipe` detection result shared between the asynchronous callback and frame processing; `self.lock` protects prediction outputs shared with the user interface.

C. Confidence Thresholding and Prediction Smoothing

A confidence threshold of 0.60 was selected through empirical testing. Thresholds of 0.40 and 0.50 produced unstable predictions with flickering between classes, while 0.70 was too restrictive and suppressed valid gestures. A sliding window of 20 frames selects the most frequent prediction within the window. Window sizes of 4 and 8 frames caused flickering, while 30 and 100 frames introduced noticeable response lag.

D. User Interface

The deployed web application interface is shown in Fig. 3. The interface is divided into two columns. The left column displays the live annotated webcam feed with an expandable gesture reference panel. The right column displays the system status, detected gesture label, corresponding emergency phrase, and a phrase history of the last five confirmed gestures. A quick actions panel provides the UK emergency number (999) and a direct link to the 999 BSL video relay service.

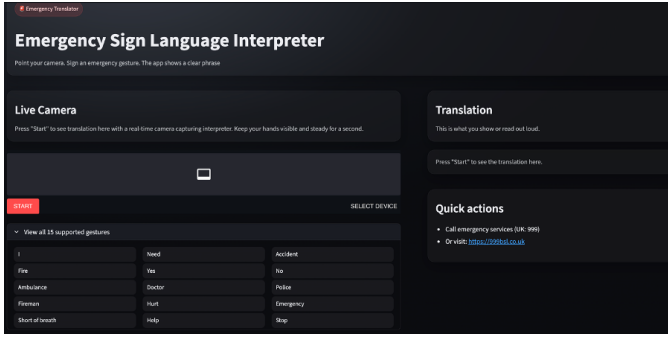


Fig. 3. Deployed web application showing live gesture detection and emergency phrase translation.

VII. EVALUATION AND RESULTS

A. Iterative Architecture Comparison

Four model configurations were evaluated to identify the best performing architecture, as shown in Table IV.

TABLE IV
MODEL ARCHITECTURE COMPARISON

Model	Architecture	Accuracy	F1
1	128-64, drop 0.2/0.3	91.67%	0.92
2	256-128, drop 0.2/0.3	92.00%	0.92
3	256-128-64, drop 0.2/0.3/0.2	92.71%	0.93
4 (final)	256-128-64, drop 0.1/0.2/0.1	95.46%	0.96

The most significant improvement came from reducing dropout rates rather than adding the third layer, suggesting the higher rates were too aggressive for the 60,000-sample dataset.

B. Training Performance

The training and validation accuracy and loss curves are shown in Fig. 4 and Fig. 5 respectively. The validation accuracy consistently exceeded the training accuracy throughout training, confirming that dropout regularisation was effective in preventing overfitting. Early stopping triggered at epoch 130, restoring the best weights with a validation loss of 0.1416.

C. Classification Performance

The final model was evaluated on 12,000 held-out test samples with 800 per gesture class. The macro average precision, recall, and F1-score were 0.96, 0.95, and 0.96 respectively. Per-class results are summarised in Table V.

D. Confusion Matrix Analysis

The confusion matrix is shown in Fig. 6. The strongest performing class was No (label 5) with a perfect precision of 1.00 and F1-score of 0.99. The lowest performing class was Hurt (label 10) with an F1-score of 0.86, showing 15 misclassifications as I and 9 as Need. This is because the Hurt gesture involves touching a body part, producing hand configurations that resemble I and Need at certain frames. Stop (label 14) shows 40 misclassifications as Help (label

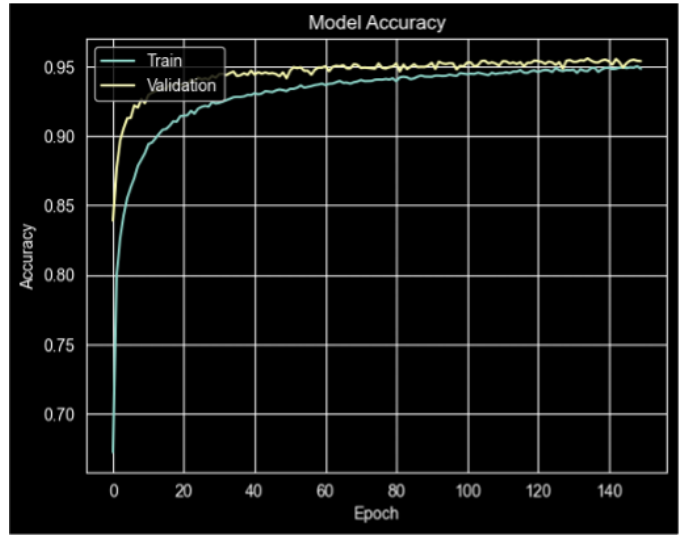


Fig. 4. Model accuracy — training and validation across epochs.

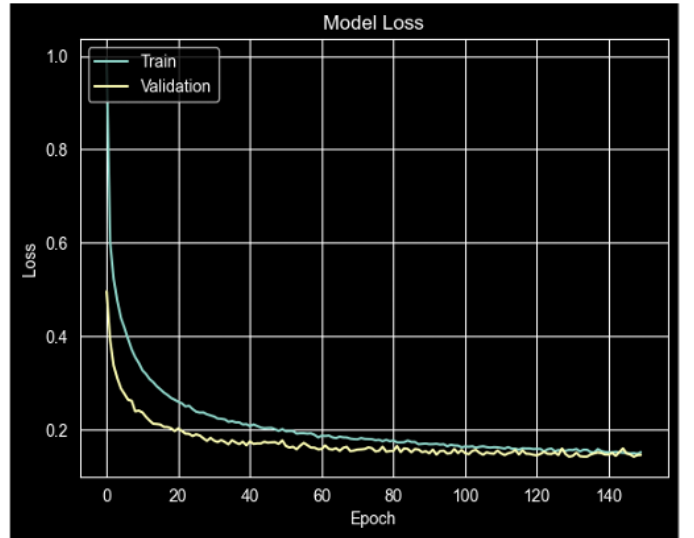


Fig. 5. Model loss — training and validation across epochs.

13), suggesting overlapping landmark configurations at certain execution points. These misclassifications are an expected consequence of per-frame classification without temporal context.

VIII. DISCUSSION AND CONCLUSION

The results demonstrate that landmark-based FNN classification is an effective and efficient approach for emergency sign language recognition. The iterative architecture experimentation revealed that dropout rate selection had a greater impact on performance than network depth alone. The system addresses the identified research gap by providing a real-time, accessible, low-cost tool bridging the communication gap between DHH and hearing individuals in emergency situations without specialist hardware.

Several limitations should be acknowledged. EmergencySL was collected by a single individual, meaning the model may

TABLE V
PER-CLASS CLASSIFICATION RESULTS

Label	Gesture	Prec.	Rec.	F1
0	I	0.95	1.00	0.97
1	Need	0.96	0.96	0.96
2	Accident	0.98	0.98	0.98
3	Fire	0.98	0.95	0.96
4	Yes	0.98	0.98	0.98
5	No	1.00	0.99	0.99
6	Ambulance	0.97	0.95	0.96
7	Doctor	0.99	0.94	0.97
8	Police	0.97	0.96	0.96
9	Fireman	0.99	0.97	0.98
10	Hurt	0.79	0.94	0.86
11	Emergency	0.97	0.98	0.97
12	Short of breath	0.97	0.92	0.94
13	Help	0.92	0.91	0.91
14	Stop	0.94	0.89	0.91
Macro avg		0.96	0.95	0.96

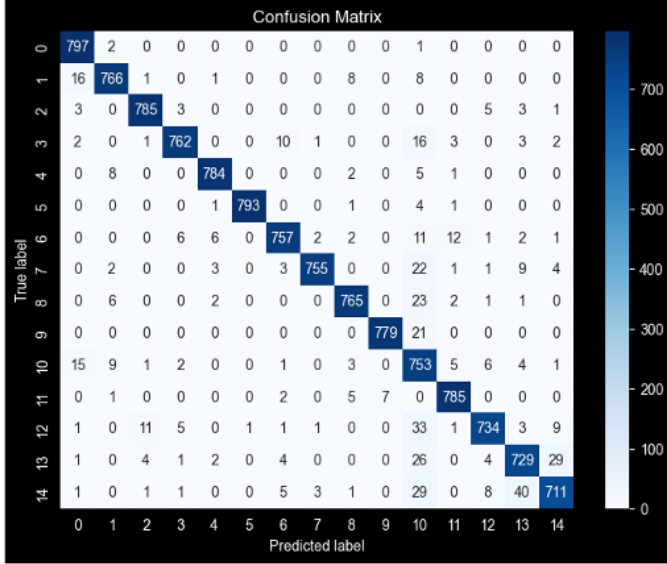


Fig. 6. Confusion matrix across all 15 gesture classes.

not generalise equally to users with significantly different signing styles. The per-frame classification approach cannot capture the temporal dynamics of dynamic gestures such as Hurt and Help. The gesture vocabulary is limited to 15 classes with phrase mapping used for demonstration purposes only.

Future work should focus on collecting data from multiple signers, restructuring EmergencySL into temporal sequences for LSTM-based training, and expanding the gesture vocabulary to cover complete emergency phrases.

In conclusion, this paper presented a real-time vision-based hand gesture recognition system for emergency sign language interpretation achieving 95.46% test accuracy and a macro F1-score of 0.96 across 15 gesture classes, running entirely on CPU-based systems without specialist equipment. The introduction of EmergencySL represents a contribution to the field with potential to support future emergency sign language

research.

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